www.Padasalai.Net www.TrbTnpsc.com

12 R

Quarterly Examination - 2025

MATHEMATICS

Time : 3.00 hrs. Reg. No. Max. Marks : 90

PART – I

Choose the correct answer 20 x 1 = 20

1. If $|adj(adj\ A)| = |A|^9$ then the order of the square matrix A is a) 3 b) 4 c) 2 d) 5
2. If $A = \begin{bmatrix} 2 & 0 \\ 1 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 4 \\ 2 & 0 \end{bmatrix}$ then $|adj\ (AB)| =$ a) −40 b) −80 c) −60 d) −20
3. If $A = \begin{bmatrix} 2 & 3 \\ 5 & -2 \end{bmatrix}$ be such that $\lambda A^{-1} = A$, then λ is.....a) 17 b) 14 c) 19 d) 21
4. If $|z| = 1$, then the value of $\frac{1+z}{1+\bar{z}}$ is a) z b) $\bar{z}$ c) $\frac{1}{z}$ d) 1
5. The principal argument of $(\sin 40^\circ + i\cos 40^\circ)^5$ is a) -110° b) -70° c) 70° d) 110°
6. The value of $\left(\frac{1+\sqrt{3}i}{1-\sqrt{3}i}\right)^{10}$ is a) $\text{cis}\frac{2\pi}{3}$ b) $\text{cis}\frac{4\pi}{3}$ c) $-\text{cis}\frac{2\pi}{3}$ d) $-\text{cis}\frac{4\pi}{3}$
7. A zero of $x^3 + 64$ is a) 0 b) 4 c) +i d) −4
8. The number of positive zeros of the polynomial $\sum_{r=0}^{n} nC_r(-1)^r x^r$ is.....a) 0 b) n c) < n d) r
9. $\tan^{-1}\left(\frac{1}{4}\right) + \tan^{-1}\left(\frac{2}{9}\right)$ is equal to a) $\frac{1}{2}\cos^{-1}\left(\frac{3}{5}\right)$ b) $\frac{1}{2}\sin^{-1}\left(\frac{3}{5}\right)$ c) $\frac{1}{2}\tan^{-1}\left(\frac{3}{5}\right)$ d) $\tan^{-1}\left(\frac{1}{2}\right)$
10. If $\sin^{-1}x + \cos^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{2}$, then x is equal to a) $\frac{1}{2}$ b) $\frac{1}{\sqrt{5}}$ c) $\frac{2}{\sqrt{5}}$ d) $\frac{\sqrt{3}}{2}$
11. The circle $x^2 + y^2 = 4x + 8y + 5$ intersect the line $3x - 4y = m$ at two distinct point if
    a) $15 < m < 65$ b) $35 < m < 85$ c) $-85 < m < -35$ d) $-35 < m < 15$
12. If $x + y = k$ is a normal to the parabola $y^2 = 12x$, then the value of k is a) 3 b) −1 c) 1 d) 9
13. The eccentricity of the ellipse $(x-3)^2 + (y-4)^2 = \frac{y^2}{9}$ is..... a) $\frac{\sqrt{3}}{2}$ b) $\frac{1}{3}$ c) $\frac{1}{3\sqrt{2}}$ d) $\frac{1}{\sqrt{3}}$
14. If $a = 1 + i$ then a^2 = ...a) $1 - i$ b) $2i$ c) 2 d) $i - 1$
15. The area of the circle $(x-2)^2 + (y-k)^2 = 25$ is a) 25π b) 5π c) 10π d) 25
16. If the line is parallel to the plane, then the line is.....to the normal to the plane.
    a) parallel b) perpendicular c) equal d) none of these
17. If $\vec{a}.\vec{b} = \vec{b}.\vec{c} = \vec{c}.\vec{a} = 0$ then the value of $[\vec{a}, \vec{b}, \vec{c}]$ is.....a) $|\vec{a}||\vec{b}||\vec{c}|$ b) $\frac{1}{3}|\vec{a}||\vec{b}||\vec{c}|$ c) 1 d) −1
18. If the direction cosines of a line are $\frac{1}{c}, \frac{1}{c}, \frac{1}{c}$ then a) $c = \pm 3$ b) $c = \pm\sqrt{3}$ c) $c > 0$ d) $0 < c < 1$
19. If the length of the perpendicular from the origin to the plane $2x + 3y + \lambda z = 1$, $\lambda > 0$ is $\frac{1}{5}$, then the value of λ is
    a) $2\sqrt{3}$ b) $3\sqrt{2}$ c) 0 d) 1
20. The value of $\sin^{-1}(\cos x)$, $0 \le x \le \pi$ is.....a) $\pi - x$ b) $x - \frac{\pi}{2}$ c) $\frac{\pi}{2} - x$ d) $x - \pi$

PART - II

Answer any seven questions. Question No.30 is compulsory. 7 x 2 = 14

21. If $adj\ A = \begin{bmatrix} -1 & 2 & 2 \\ 1 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$, find A^{-1}.
22. Find the rank of Matrix : $\begin{bmatrix} -1 & -2 & 3 \\ -2 & 4 & 6 \\ 5 & 1 & -1 \end{bmatrix}$
23. Show that $|3z - 6 + 12i| = 8$ represents a circle and find its centre and radius.
24. Which one of the points i, −2 + i, and 3 is farthest from the origin?
25. If α, β, γ are the roots of the equation $x^3 + px^2 + qx + r = 0$ find the value of $\sum\frac{1}{\beta\gamma}$ in terms of the coefficients.

Scanned with Cam
Kindly Send Me Your Study Materials To Us Email ID: padasalai.net@gmail.com

www.Padasalai.Net www.TrbTnpsc.com

26. Find the principal value of $\sin^{-1}(2)$, if its exists.

27. Find the value : $\tan^{-1}\left(\tan\frac{5\pi}{4}\right)$

28. Find the general equation of a circle with centre (–3, –4) and radius 3 units.

29. Show that the vectors $\vec{i}+2\vec{j}-3\vec{k}, 2\vec{i}-\vec{j}+2\vec{k}$ and $3\vec{i}+\vec{j}-\vec{k}$ are coplanar.

30. Show that $[\vec{a}-\vec{b}, \vec{b}-\vec{c}, \vec{c}-\vec{a}]=0$

PART - III

Answer any seven questions. Question No.40 is compulsory. 7 x 3 = 21

31. Solve the system of linear equations by Cramer's rule : $5x - 2y + 16 = 0$, $x + 3y - 7 = 0$

32. Simplify : $\left(\frac{1+i}{1-i}\right)^3 - \left(\frac{1-i}{1+i}\right)^3$ into rectangular form.

33. Find the value of $\left[\frac{1+\sin\frac{\pi}{10}+i\cos\frac{\pi}{10}}{1+\sin\frac{\pi}{10}-i\cos\frac{\pi}{10}}\right]^{10}$

34. Show that the polynomial $9x^9 + 2x^5 - x^4 - 7x^2 + 2$ has atleast six imaginary roots.

35. Find the value of $\cos^{-1}\left(\cos\frac{\pi}{7}\cos\frac{\pi}{17} - \sin\frac{\pi}{7}\sin\frac{\pi}{17}\right)$

36. Show that $\cot^{-1}\left(\frac{1}{\sqrt{x^2-1}}\right) = \sec^{-1}x$, $|x| > 1$.

37. A concrete bridge is designed as a parabolic arch. The road over bridge is 40m long and the maximum height of the arch is 15m. Write the equation of the parabolic arch.

38. Find the shortest distance between the two given straight lines $\vec{r}=(2\vec{i}+3\vec{j}+4\vec{k})+t(-2\vec{i}+\vec{j}-2\vec{k})$ and $\frac{x-3}{2}=\frac{y}{-1}=\frac{z+2}{2}$

39. Find the angle between the straight line $\vec{r}=(2\vec{i}+3\vec{j}+\vec{k})+t(\vec{i}-\vec{j}+\vec{k})$ and the plane $2x - y + z = 5$.

50. If A is orthogonal and $A^T = \begin{bmatrix} 2 & -3 \\ -1 & 1 \end{bmatrix}$ then find A.

PART - IV

Answer all the questions 7 x 5 = 35

41. a) Investigate for what values of λ and μ the system of linear equations $x + 2y + z = 7$, $x + y + \lambda z = \mu$, $x + 3y - 5z = 5$ has (i) no solution (ii) a unique solution (iii) an infinite number of solutions **(OR)**

b) Find the vertex, focus, equation of directrix and length of the latrus rectum of $x^2 - 2x + 8y + 17 = 0$.

42. a) If $z = x + iy$ and $\arg\left(\frac{z-i}{z+2}\right)=\frac{\pi}{4}$ show that $x^2 + y^2 + 3x - 3y + 2 = 0$ **(OR)**

b) If $\tan^{-1}x + \tan^{-1}y + \tan^{-1}z = \pi$, show that $x + y + z = xyz$.

43. a) If the roots of $x^3 + px^2 + qx + r = 0$ are in H.P. prove that $9pqr = 27r^2 + 2q^3$. **(OR)**

b) Using vector method, prove that $\cos(\alpha - \beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta$.

44. a) Solve the equation $z^3 + 8i = 0$, where $z \in C$. **(OR)**

b) A rod of length 1.2 m moves with its ends always touching the co-ordinate axes. The locus of a point P on the rod, which is 0.3 m from the end in contact with x axis is an ellipse. Find the eccentricity.

45. a) If $A=\frac{1}{9}\begin{bmatrix} -8 & 1 & 4 \\ 4 & 4 & 7 \\ 1 & -8 & 4 \end{bmatrix}$ prove that $A^{-1} = A^T$. **(OR)**

b) Show that the lines $\frac{x-3}{3}=\frac{y-3}{-1}$, $z - 1 = 0$ and $\frac{x-6}{2}=\frac{z-1}{3}$, $y - 2 = 0$ intersect. Also find the point of intersection.

46. a) Find the equation of the circle passing through the points (1, 0), (1, 1) and (2, 1) **(OR)**

b) Find the parametric form of vector equation and Cartesian equations of the plane containing the line $\vec{r}=(\vec{i}-\vec{j}+3\vec{k})+t(2\vec{i}-\vec{j}+4\vec{k})$ and perpendicular to plane $\vec{r}.(\vec{i}+2\vec{j}+\vec{k})=8$.

47. a) Solve the equation $6x^4 - 5x^3 - 38x^2 - 5x + 6 = 0$ if it is known that $\frac{1}{3}$ is a solution. **(OR)**

b) If $\cos^{-1}x + \cos^{-1}y + \cos^{-1}z = \pi$ and $0 < x, y, z < 1$, show that $x^2 + y^2 + z^2 + 2xyz = 1$.

Scanned with Cam
Kindly Send Me Your Study Materials To Us Email ID: padasalai.net@gmail.com